

Introduction to AI Agents

Agenda

- LLMs vs AI Agents
- Comparing AI Agent Architectures
- Static vs Dynamic Environments
- Groundedness and Validation in AI Agents

Let's begin the discussion by answering a few questions.

Introduction to AI Agents Quiz

What is the key difference between an LLM (Large Language Model) and an AI Agent?

A

LLMs can only process numbers, while Agents can process text.

B

LLMs generate outputs based on prompts, while Agents use LLMs along with memory, planning, and tools to take actions.

C

Agents are trained models, while LLMs are just databases.

D

There is no difference; LLMs and Agents are the same.

Introduction to AI Agents Quiz

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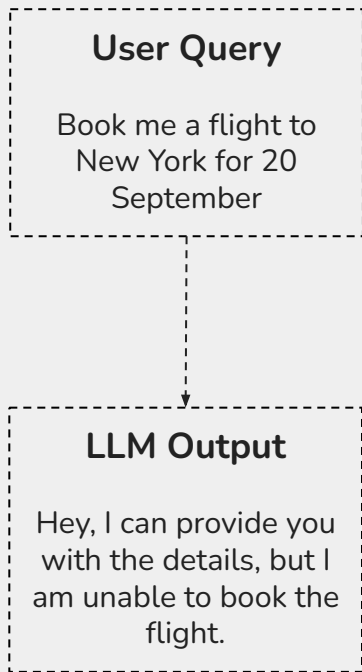
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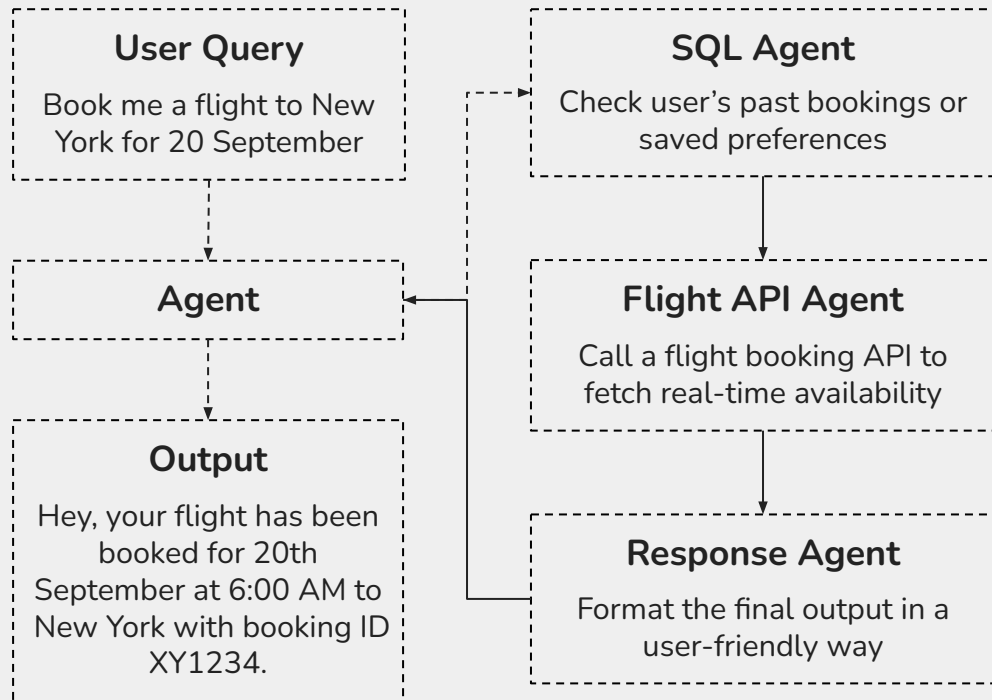
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LLM vs Agents

LLM



AI Agents



Introduction to AI Agents Quiz

A navigation system continuously monitors the user's position and, upon detecting deviation from the planned route, immediately recomputes an alternative path without consulting the user. This behavior is best classified as:

A

Interactive agent behavior

B

Proactive agent behavior

C

Reactive agent behavior

D

Non-autonomous behavior

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Comparing Agent Architectures

Feature	Reactive Agents	Proactive Agents	Interactive Agents
Core Behavior	Responds to current inputs without explicit planning or future state reasoning	Anticipates needs and plans ahead	Engages in dynamic, two-way dialogue
Decision Logic	If-then rules or learned policies that do not involve look-ahead or predictive planning	Goal-oriented with explicit planning and predictive reasoning	Conversational policies that track user intent and context
Memory	Typically stateless(or very limited state in simple reflex-with-state variants)	Uses historical data and an internal model of the environment	Maintains session and user context across turns
Initiative	User or environment triggered	Self-initiated actions driven by goals and forecasts	User- or agent-initiated; emphasizes collaboration and engagement
Example	FAQ Assistant	Reminder assistant	Support Chatbot

Comparing Agent Architectures

Reactive Agents

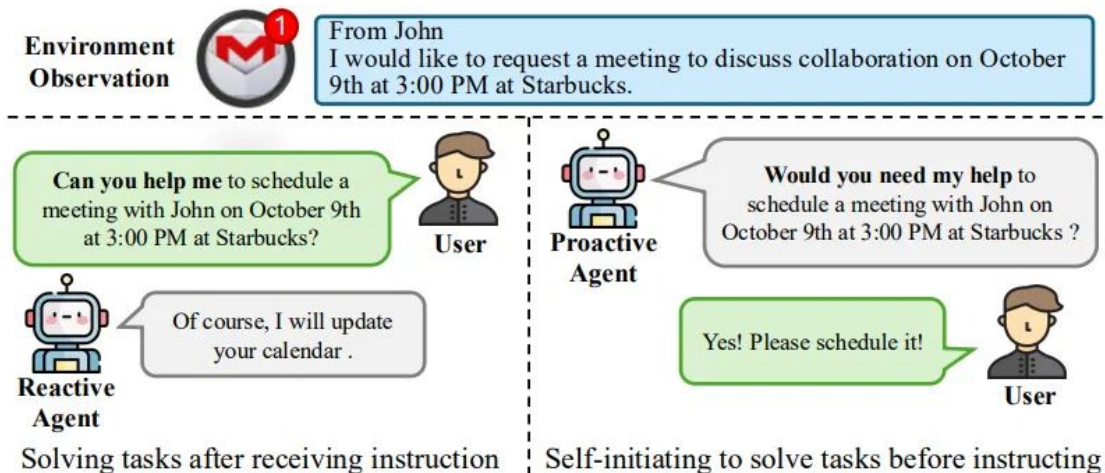
The Foundation: High speed, low complexity. Best for stable environments where immediate response is critical.

Proactive Agents

The Strategist: Medium complexity, high autonomy. Best for optimizing future outcomes and reducing user friction

Interactive Agents

The Partner: High complexity, high engagement. Best for tasks requiring empathy, persuasion, and complex communication.



Introduction to AI Agents Quiz

Which condition correctly classifies an environment as dynamic for an agent?

A

The environment can change while the agent is reasoning or acting

B

The agent acts only once and then stops

C

The same action always gives the same result

D

The agent has complete knowledge of the environment

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Static vs Dynamic Environments

Aspect	Dynamic Environment	Static Environment
Environment change	Can change while the agent is reasoning or acting	Does not change while the agent is deliberating
Time sensitivity	Time matters; delays can change the state or reduce performance	Time does not affect the environment itself (performance may still depend on computation time in practice)
Agent requirement	Should continuously observe and adapt (continual sensing or frequent updates required)	Can deliberate and then act without continual sensing during deliberation
Examples	Traffic systems, live markets, real-time user interactions	Chess (during a single move), crossword puzzles

Introduction to AI Agents Quiz

Why are groundedness and validation necessary in an agentic system?

A

To increase creativity

B

To reduce latency

C

To ensure evidence-based and rule-safe outputs

D

To avoid tools and external data

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Groundedness and Validation in Agents

Aspect	Groundedness	Validation
Purpose	Ensures responses are based on verifiable external knowledge sources	Ensures responses meet quality, correctness, safety, and policy constraints
Focus	Preventing hallucinations and unsupported claims	Preventing incorrect, unsafe, or non-compliant outputs
How it works	Grounds the agent on documents, databases, or tool outputs (e.g., RAG)	Applies rule-based checks or secondary evaluation models
When applied	During reasoning and response generation	Before final response delivery
Why it matters	Builds trust and factual accuracy	Ensures compliance, reliability, and consistency
Examples	Knowledge retrieval, enterprise QA, decision support	Regulated domains, production AI systems, multi-agent workflows



Power Ahead!

